

LINUX COMMANDS

Experiment NO. 2

Date : 6/09/25

Rollno : B019

Aim : To perform basic commands of linux.

Theory : linux is an operating system on kernel. It is distributed under an open source license. It's list is quite like UNIX. linux O.S germinated as an idea is the mind of young & bright Linus Torvald's when he was a computer science student. The main advantage of linux is that the programmers are able to use the linux kernel to design their own custom operation system. linux comprises of 2 model :

① Kernel mode ② user mode

- Kernel mode, kernel component code executes in a special privileged mode called kernel mode with full access to all resources of the computers.
- user mode : support code which is not required to run in kernel mode is in system library. user programs and other system programs work in user mode which have no access to hardware.

Basic feature of linux O.S are as follows :

- 1] Portable
- 2] Open source
- 3] Multi-programming
- 4] Hierarchical file system

- 5] Multi user
- 6] shell
- 7] security
- 8] Hardware
- 9] kernel
- 10] utilities

Procedure : Basic unix commands

- ① Pwd : print working directory
- ② cal : print the calendar of the current month
- ③ cal mont no. year
eg: cal 08 2002
→ August 2002
- ④ echo → echo whatever you provide
eg: echo "Hello world"
O/P :- Hello world
- ⑤ date command : Display current time & date
eg: date
O/P : Fri sept 6 2:30:55 IST 2019
→ If we want to display any time then date +I.I
command
O/P - 2:30:56
- ⑥ whoami : This command displays who is currently logged in the terminal.
- ⑦ clear : clears up the command prompt screen.

- ⑧ **help** : This command shows the usage summary of the command and needing it.
- ⑨ **what is** : This command gives 1 line description of any command.
- ⑩ **is** : This command lists of files given in the directory the director name should be given after is. eg : is my folder.
- ⑪ **mkdir** : This command states a directory
- ⑫ **touch** : used for creating empty file.
- ⑬ **rmdir** : removes directory name
- ⑭ **copy** : copies files on directories
- ⑮ **move or mv** : moves files or directories. This command works like CP command. The difference is that the original file is removed from source to destination.
- ⑯ **head** : This command displays first lines at a file
- ⑰ **tail** : This command displays last lines at a file.

⑱ wc : This command gives word count of files.
The output is no. of files, no. of words,
no. of letters.

⑲ vi : stands for visual editor. It is a text editor in linux. we can use the editor to open a file & edit it.

⑳ history : displays how many commands inserted in your terminal.

conclusion: Hereby we have learned to perform basic linux command.

Pranav
6/11/20

WINDOWS
OS



SVKM'S
SHRI BHAGUBHAI MARATHI POLYTECHNIC & COLLEGE
OF ENGINEERING
IN A HIGHER EDUCATION INSTITUTION



Course: Computer Engineering

Sem: 1 (B)

LIST OF EXPERIMENTS IN FUNDAMENTALS OF COMPUTING SYSTEM: EC5238105

Sr No	Title	Page No.	Date	Remark
1.	To perform network commands and view network architecture	1-7	30/08/25	✓ A.P.A.
2.	To perform basic commands of linux	8-12	6/09/25	✓ A.P.A.
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4.	To create tables & mail merge documents in microsoft word	21-30	20/09/25	✓ A.P.A.
5.	To perform various functions of word like replace, go to, page setup, print preview print custom doc, macros filter etc	31-36	21/09/25	✓ A.P.A.
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7A.	To learn basic features of spreadsheets	46-49	18/10/25	✓ A.P.A.
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This is to certify that the Shri Bha student ghavari Roll No. B019 has performed the above mentioned experiments in the laboratory of Shri Bhagubhai Marathi Polytechnic of College of Engineering during the year 2025-2026

Signature

Head of Department

Signature
10/10/25

Signature
10/10/25

WINDOWS
OS



**SHRI BHAGUBHAI MAFATLAL POLYTECHNIC & COLLEGE
OF ENGINEERING**



Computer Engineering

IS OF EXPERIMENTS IN FUNDAMENTALS OF COMPUTING SYSTEM FC5238905

[illegible]

Roll No. 5019

and the above mentioned experiments in the laboratory of Shri Bhagubhai Mafatlal Polytechnic of Engineering during the year 2025-2026

Principal

Experiment no. 03

Date: 13/09/2025

Roll no: B019

Aim: To get familiar with different means of windows.

Theory: Windows 0-5 is the one of the most commonly used

0-5. It contains some basic elements. These basic elements can be grouped into 2 types

1. System configuration element.
2. UI element.

The system configuration elements are as follows:

1.1 Control panel:

gives the user the ability to change basic system settings. users may also add hardware components as well as add/remark software.

1.2 Device manager:

gives the user the ability to display and control computer hardware.

1.3 windows mobility:

Provides the user with information related to mobile computing.

1.4 security and maintenance:

Provides the user with information related to security components within the operating system.

- The UI elements are

2.1 windows shell:

Provides the main mechanism through which the window GUI is presented. All dialogue boxes, menus, taskbar, start button menu and windows explorer are contained within the windows shell.

2.2 windows explorer:

gives the user access to file and software operation while allowing the user to launch applications and perform tasks.

2.3 File Folder:

Provides user with a visual way in which to view files that are contained, stored in a particular location.

2.4 windows search:

Provides user with a way to explore all files on hard disk.

2.5 start menu:

The launch point for all software applications location into the windows O.S. All software appear in this menu as a resulted list.

2.6 Task Bar:

Application bar used to launch and monitor software applications.

2.7 Icons :

Small graphical representation of a program or a file that when clicked on will be run or opened.

2.8

1) Word processing : This class of software provides for the creation of written document. Microsoft's word processing application is Microsoft word.

2) Presentation : This class of software provides for the creation of slideshow presentation which can include table, charts, images & text. Typically, presentation software allows the user to include notes on slides that are not visible to viewers.

3) Spreadsheet : This class of software provides away to do numeric calculation and analysis. The working area and divided into rows and columns. Microsoft spreadsheet application is Microsoft excel.

4) Graphics : This class of software provides for the user to create pictures, diagrams, charts and graphs.

5) Electronic Mail : This class of software provides for the user for sending and receiving email messages.

7] Start menu : when a menu appears with an arrow this means that menu item holds its own list called as sub menu. To access the menu simply point the mouse cursor on the menu item has arrow.

⇒ Windows menu :-

To use a menu you must first find whether and where it is available. The menu that appears depends on what you have just clicked. The taskbar, an empty area on the desktop or picture on desktop.

⇒ opening a menu :-

1. Start the computer
2. Click an empty area on the desktop and notice that nothing happens and nothing displays.
3. On the taskbar, click start.
4. Notice that a menu appears.
5. Right click area on desktop. Notice a different menu appears.
6. Right click an empty area on Taskbar and a different menu appears.
7. Click an empty area on desktop. Notice that this menu dismisses any menu that appears.

⇒ Menu categories : To diversity the actions that can be performed on a computer, there are 6 different main categories of menu :-

Pran

— x — x — x —

we have learnt different menus of window 0.5.
Conclusion :

8. To dismiss the menu click start again.
7. Notice that many names of the program appears programs.
6. On the taskbar, click start position the mouse cursor on top left corner.
5. To close the window, double click the small picture on its
4. On the menu, click explore.
- type of menu appears.
3. On the taskbar, rightclick start and notice that a different
2. To dismiss the menu, click an empty area on the desktop.
1. Open Taskbar, click start.

⇒ Using categories of menu :-

1] Arrow menu :- when a menu appears with an arrow this means that menu item holds its own list called a sub menu. To access the menu simply position the mouse cursor on the menu item has arrow.

Adjust your computer's settings

View all Category ▾



System and Security

Review your computer's status
Save backup copies of your files with File History
Backup and Restore (Windows 7)



Network and Internet

View network status and tasks



Hardware and Sound

View devices and printers
Add a device



Programs

Uninstall a program
Get programs



User Accounts

Change account type



Appearance and Personalization



Clock and Region

Change date, time, or number formats



Ease of Access

Let Windows suggest settings
Optimize visual display

WCJJD
PCKGES

Experiment no. 04

(A)

: working with word Packages

Date: 20/09/2025

Rollno: B019

Aim: To perform basic features of word packages.Theory:

word processing software application developed by microsoft. This enables you to create simple and complex documents such as reports and letters. We can change the content of documents such as spellcheck, formatting the content, putting different style and edit the content.

Procedure:

1. Create a folder of your name and create a subfolder of FCS.
2. Create a blank document called as Experiment no. 04.
3. Write 3 paragraphs with content about your personal details, about your school details and about your future details and plans.
4. Use different types in formatting toolbar.
eg. left text, right text, justified text, centered text.
5. Use different alignment for paragraphs.
eg. Font size, font style, etc.
6. Use different text effect for paragraphs.
(Home tab → Font group → text effect)
eg. outline, shadow, reflection, glow, etc.

autosave  

Document 2 - Mode



File Home Insert design layout references mailings review view help



Paste



clipboard

Calibri v 11 v

B I U v  x2 x2

A v  v A v A v

Paragraph



AaBbCc
Normal



editing



dictate
voice

styles

7. Justify all the paragraph within the margin (page layout)
8. Clear all the style applied to the paragraph (Ctrl + A) (to select text and click on clear all formatting button)
9. Apply bullet style to all the paragraphs.
(Home > paragraph > click arrow next to bullets)
10. Change the indent of all the paragraphs.
(select the paragraph then go to page layout (layout tab) in paragraph group. choose paragraph settings, choose indentations and spacing tab & choose your setting and click on)
11. Apply border styles and different styles and different paragraphs. (paragraph to t > borders > select your choice)
12. Give different columns to different words in each paragraph and highlight specific words in each paragraph (select word → font > select your colour choice)
13. Apply borders and shading to third paragraph (paragraph group → click arrow next to border > borders & shading)
14. Paragraph edit options: cut, copy, paste, find, replace (clip board group < Home tab)
15. Use autocorrect option for the whole content of the document (File - option - proofing - auto correct)
16. Insert date and time in the document.
(insert tab > text group > date > time > insert your choice)
17. Insert page break after each paragraph.
(go to layout > breaks > page)

18. Insert page no. and change the margins of the page using ruler (Insert > header & footer > page number > page layout > page setup > margins > select your choice)
19. View the document in following layout (view)
 - a) print layout b) web layout c) online layout
 - d) full screen view > view group
20. Protect your document with password at your home.
(file → info → protect document)
21. A. Transfer first paragraph to french
B. Transfer second paragraph to hindi
C. Transfer third paragraph to Arabic. (File option language)
22. Create the following bulleted list in some document.
Apple : ☐ red ☐ himachal
 grapes ☐ green ☐ black
 orange : ☐ Niagara ☐ Fedorera
 Helen : ☐ musk ☐ water
23. Insert line no. to each line (page layout - line no.)
page layout > page setup > line number > select option
24. Convert document into 2 columns (page layout - column)
page layout > page setup > column
25. Insert a picture in document
(Insert > illustrations > pictures / picture online > select picture you want to insert)
26. Wrap your text around the text
(select the picture, select format then under arrange select wrap text. Choose wrapping option you want)
27. Work with option underline and give different style to your text. Insert > Text group > underline

28. Insert a clip art and select pictures > picture online > online pictures then type word/phrase to describe what you are looking for, then press enter. Filter the results by type for clipart, select a picture, select insert.
29. Select rotational diagram (select object to rotate > shape on picture format > conclude > rotate).

Conclusion : Learnt about different functions and formatting and editing tools in word package.

Shree
30/9/25

MAIL
MERGE

(8)

Date: 20/09/25 8019

Aim: To create tables and mail merge documents in microsoft word.

Theory:

Tables are used to represent tabular format in document to create a table. We can choose insert option and select no. of columns and rows. The column width can be set to auto which means the width of column can be decided by contents in it.

Mail merge: is a handy feature that incorporates data from both microsoft word and microsoft excel and allows you to create multiple documents at once such as letters, saving your time & effort of retyping the same letter over and over.

Procedure:

Table 1

Name	Marks			Total	Percentage
	Maths	Science	English		
ABC	10	10	10	30	100%
PQR	5	6	7	18	60%
XYZ	9	6	7	22	73.33%.

1. Calculate the total & percentage (layout-formula) using equation.

2. Insert a column of roll no to the left of student name.

- (to insert a column to the left of cell-click insert left in the rows and columns group)
- 3 Add column GRADE on right side of percentage. (To add column to right of cell click insert height in rows & columns group)
- 4 Insert caption for table (click on figure/table where you want caption to appear on reference tab, click the insert.
- 5 start the table content according to percentage and according to the name of students (select anywhere in table → select table tools layout > sort > select any criteria > select OK)
- 6 Add a row at the end of table to add a new student
- 7 Add a new row between 1st and 2nd add new student's details
- 8 Change of column headers to upper case to lower case Home tab-change case in font group (A) - a drop down menu appears - select desired case
- 9 Give border to the table. (select table border and style)
- 10 Insert an auto shape below table (insert > shape)
- 11 Use eraser & erase some area of table. (layout > eraser)

Mail merge

- * write a letter to your letter inviting him/her to attend your friend birthday party

<< First name >> << last name >>

<< mailing address street >>

<< mailing address city >> << mailing address state >>

<< mailing address : >>

Dear << Friend name >>

sample letter | template for mail merge feature instructions.

Sincerely,

Wealthbox Team

* step by step procedure to write mail merge

- click the mailings tab
- click the start mail merge button
- select step by step mail merge wizard
- select the type of document to create
- click next : starting document

* select a document

- select a starting document
- You can use the current document as the basis for the mail merge
- click next : select recipients
- select use an existing file.

- click browse
- select your data file
- click open
- Move sure right recipients are selected and click OK
- click next : write your letter

* write your letter

- click where you want information
- select one of placeholder options
- Address block : This is a combination of fields to insert the name & addresses of recipients

greeting line : This is a combination of fields to insert the recipients name in the greeting line

insert mail merge : when you click this button, a list of additional merge fields you can insert appears.

- customise the placeholder
- click OK
- (optional) repeat steps to add each merge field you want to include
- click next : Preview your letters

Conclusion :

Learned various implementing functions on table & how to use mail merge.

12/10/2025

FUNCTIONS OF WORDS

Experiment No. 05Date: 27-09-25Roll no: B019

Aim: To perform various functions of word like find replace go to, page setup, print preview, print custom sort, macros, filters, etc.

Theory:Find and replace

- In the "Home" menu bar, click on "Replace" in the "Editing" section or use the keyword shortcut "Ctrl + H".
- You can then choose to replace one at a time by clicking on "Find Next" then "Replace" each time or click "Replace All" to replace all at once. Word will quickly scan the document and you can hit "OK" when its finished.

Advanced features in word's find & replace tool

- In the Home menu bar, click replace in the editing section or use keyboard shortcut Ctrl + H.
- In the box next to Find what, type 2 spaces by hitting the space bar twice.

- In the box next to replace with type 1 space by hitting the space bar once. You can see in the example below that it has found two spaces between "how" & "you".
- You can then choose to replace one at a time by clicking Find Next then Replace.

Go to :

- From the Home Tab in the Editing group select Find & then select go to.
- On the GO TO tab of the Find & Replace dialogue box use the options under .go to what to locate certain elements.
- In this text box, enter plus (+) or minus (-) a number to move to move through the specified element in the document & click go to.

Page setup

Paper size

- 1 On the File menu, click Page setup. The Page setup dialog box is displayed.
- 2 On the page setup dialog box, click the paper tab.

3 choose appropriate option in the apply box & click ok.

4 Margins

1. On the File menu, click Page setup & then Margins tab.
2. In the top, bottom, left & right boxes, enter a desired measurement for the respective margin & click OK.

Print Preview

Print preview is a feature that displays on the screen. Choose File → Print preview from menu bar. Zoom in & take a look at the print preview toolbar.

- The print button prints a file.
- The view Ruler button displays the vertical & horizontal rulers. Rulers help edit margin settings.
- If your file consists of multiple pages, the multiple pages button allows you to either view one page or several pages at a time.
- Use zoom, then use the scroll bars to view your file.
- Use the shrink to fit button to fit your information to one page.
- If you want to exit or close print preview, click close close on 'Print preview toolbar'.

Print your work

Choose File → Print from the menu bar (Ctrl + P)

Sort data using a custom list

with built-in custom lists, you can sort data - either by days of the week or months of the year. optionally, you can create your own custom lists to sort by any other characteristics that doesn't sort well alphabetically such as high, medium and low - or S, M, L, XL

sort by days of the week or months

Follow these steps

1. select the columns to sort. In the ribbon Data > sort. In the sort pop up window, in the sort of drop-down choose the column on which you need to sort.

For example, if you want to re-sort the previous examples by delivery data used sort by choose "delivery" from the order drop-down.

select custom list. In custom lists box, select the list that you want then click OK to sort the worksheet.

create your own custom list

Follow these steps to create your own custom list for sorting:

1. In a column of a worksheet, type values to sort by. Arrange them in the order that you want to define the sort order - from top to bottom. For example.

	A
1	High
2	Medium
3	Low

1. select all of the cells in that list, and then clicks **File > options > Advanced**.

2. scroll down to **general** section & click **edit custom list** in the custom list box, click **Import**.

Filter data in a range or table

Filter a range of data

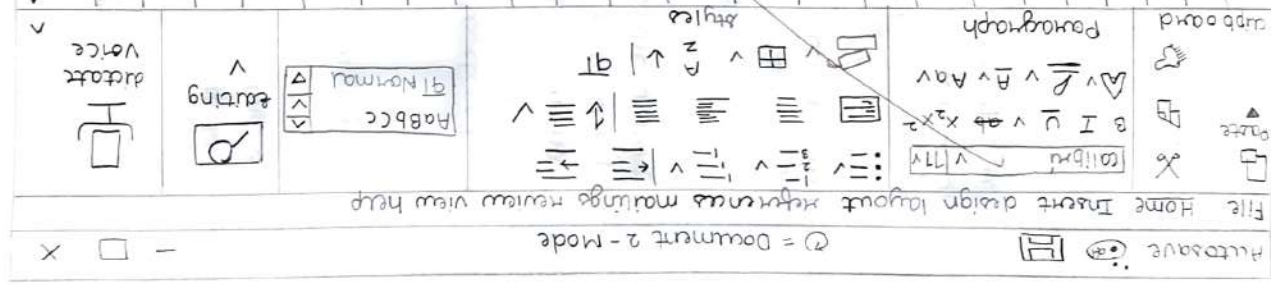
- select any cell within the range
- select **Data > Filter**
- select the column header arrow ☒
- select **Text filter** or **Number filters** & then select a comparison, like **Between**.
- enter the filter criteria & select **OK**.

Conclusion: learnt to implement various functions of words like **Find & Replace**, **Page setup**, **Print preview**, **print**, **Custom sort**, etc.

Pran
27/9/25

MICROSOFT

POWERPOINT



Experiment No 06

Working with powerpoint

Date: 4/10/25

Roll no: 6019

Aim: To learn basic function of PPT

Theory: A powerpoint slideshow (ppt) is a presentation created in software from Microsoft that allows user to add audio, visual and audio/visual features to a presentation. It is considered to be a multimedia technology and also acts as a tool for collaboration and content sharing. Powerpoint is included in Microsoft Office, making it one of the most well known & widely used brands of presentation software.

Procedure:

1) Create PPT for your exp

Insert 5 slides choose a layout colouring and design according to your creativity

Slide 1 - contain healthy diet

Slide 2 - different steps towards healthy diet

Slide 3 - different strategies for healthy diet

Slide 4 - different tips for cooking food

Slide 5 - Insert various links which you have used.

2) Create different types of list (Bulleted list, numbering, multilevel bulleted list which is used for both numbering and bulleted list) Hence > arrow next to either bullets or numbering button

3) Apply different effects to the slides
(Insert your name and stream with semester and use different types of word art in your details)
Select the slide → View → animation for effects

4) Apply custom animation to the next inside the slide
Add some images to the slide

(In windows 10 it is animation panel which you will get at right side of your window once you clicked an animation) Click where you want to insert picture on your slide. On insert tab → image group → click pic → click this device → browse the pictures you want to insert → click that picture → click insert

5) Insert a textbox in one of the slides and add some content in the textbox change the font type size and the colour. Go to Home tab under insert click text. On pop-up menu click textbox. On the slide click location of where you want to add the text.

6) Insert smart art in one of the slides.
Insert tab → illustrating group → smart art

7) Insert an audio file and audio should automatically play on all slides in normal view click the slide you want to add audio to. On insert tab → media group → audio under audio → click audio from file or clipart audio, locate & select audio clip you want & click insert. On audio icon on the slide. On the audio toolbar playback tab in audio option group, select in click sequence or automatically.



9) Insert new file video

9) Use the outline view of presentation and copy the content of slides in word document

Home tab → arrow next to new slide → click but line
ctrl+c for copy and in word Home tab → clip board
click arrow next to paste & paste

10) Apply transitions on all slides and modify speed from moderate to fast

- select the slide you want to add transition to

- select the transitions tab and decrease a transition & see its preview

- select the effects option to choose the direction and nature of transition

- select preview see what transition looks like

- To remove transition, select transition > none

11) Add commands in each slide

select slide > review > no comment or insert > comment

12) Create photo album & save it with your full name

13) Expand & collapse the outline view in slide sorter view

Conclusion: Learned basic functions of ppt

Handwritten signature

VEDA MILIND CHARAT

Presented by:

Subject:

FCS238905



Presented by:

VEDA MILIND GHARAT

Subject:

FCS238905

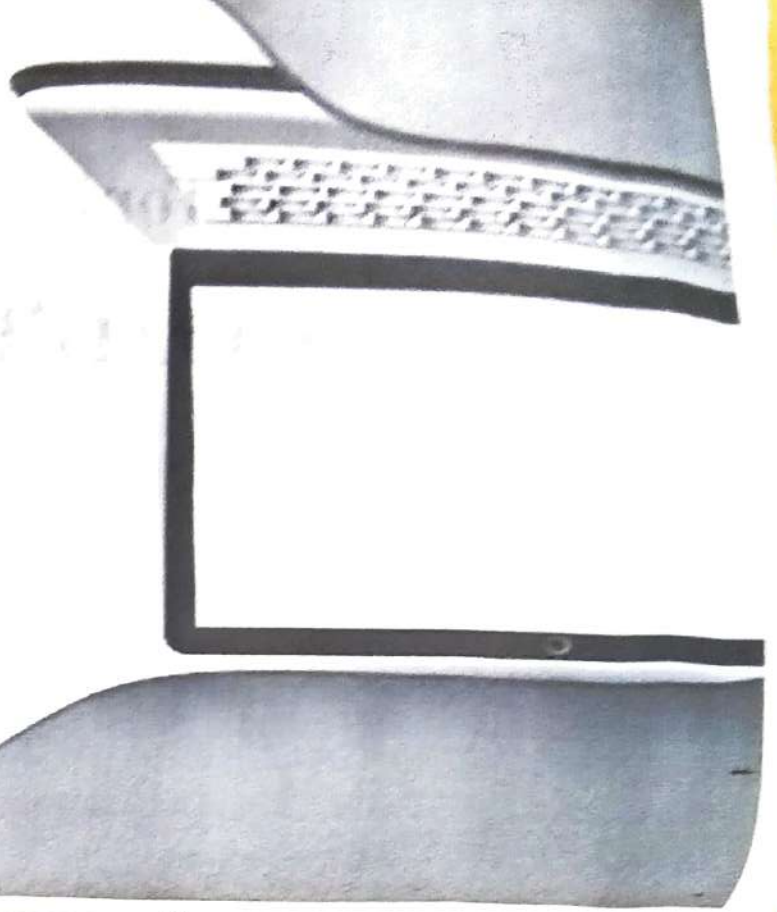


EVOLUTION OF COMPUTER



WHAT IS COMPUTER

- An electronic device that processes data to perform tasks by following programmed instructions.
- It takes data as input, processes it using a Central Processing Unit (CPU), stores information in internal memory, and then produces results through devices like screens or printers.





ADVANTAGES OF COMPI

- High Speed
- Accuracy
- Storage
- Automation
- Multitasking
- Cost Efficiency



IMPORTANCE OF COMPUTER

- Education & Learning
- Business & Work
- Communication
- Daily Life & Entertainment
- Research & Innovation

Global Connectivity

Healthcare

Global Connectivity

USES OF COMPUTER

Banking

Government & Security

Online Services

With continuous advancements like AI future of computers will be more powerful and innovative.

They connect the whole world through technology.

They make our work faster, easier, and more efficient.

Computers have become an essential part of modern life.

CONCLUSION



SPREAD
SHEETS

Experiment No. 01

(A)

Working with spreadsheet

Date: 18/10/25

Roll no: 5019

Aim: To learn basic features of spreadsheets.

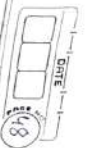
Theory:

WORKBOOKS are files created in Excel. Each workbook is created with 5 blank worksheets. WORKSHEETS are like pages within each workbook. It is where you enter, edit and edit data, worksheets are made up of columns and rows. There are 256 columns and 65,536 rows for each worksheet. A CELL is where a column and row intersect. An active cell is outline in a darker border. The formula bar is a bar located above the worksheet frame. As you enter data in a cell it simultaneously appears in the formula bar.

Procedure:

Step 1] Resizing a row/column

Hover your mouse over the border between the column or row labels. Then click and drag the border to widen/narrow the size of each cell. The columns to right or rows below will adjust to make room for your changes.



step 2) Formatting a cell

To change NUMBER FORMATTING, select cell you want to modify, click drop down arrow next to format commands on Home Tab, The number formatting drop down will appear select the desired formatting option

step 3) Inserting rows and columns

Select the row above column to the right of where the insertion should occur or click arrow to right of in row.

Choose home $\rightarrow$ cells and click arrow to right of insert button to open the drop down list for insert button. From the menu, choose insert sheet rows / insert sheets columns

step 4) Use of autosum fraction

Select a cell next to numbers you want to sum, click Autosum on Home tab, press enter and you're done.

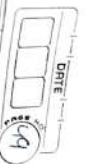
step 5) To calculate percentage

When you want to display percentage in cell use the percentage number format [Home tab $\rightarrow$ number (ribbon) $\rightarrow$ percentage] which will automatically display a decimal value as percentage.

The following formula can be used = (C6/C11) * 100

step 6) To calculate average

Click a cell below or right of the to which you want



to find average On the Home Tab in editing group, click review next to Autsum, click average and then press Enter.

step 7) Deleting rows and columns .

select rows and columns you want to delete choose Home $\rightarrow$ cells $\rightarrow$ delete .

step 8) To calculate status of students
you can use the formula.
If (F7 > 40, "Pass FAIL")

step 9) inserting a chart

select data that you wish to display as a chart. on the insert tab in charts group, select type of chart you wish to insert by clicking on it .

step 10) Automating

select the data in the worksheet that you want to format. go to quick access toolbar and select auto format. In the autformat dialog box choose a style select OK to choose the dialogue box. The new style is applied to table .

Conclusion: Here we learnt, the basic functions of spreadsheets .

Shalini
13/11/25

PROJECT TAXILES

Experiment No 07

(B)

Pivot table in excel

Date: 18/10/25

Rollno: 8019

Aim: To implement pivot tables in excel.

Theory: Pivot tables are used to represent information in expert. We can view summarized and detailed report using pivot table. Pivot table is a tool that allows user to extract answers to basic questions about data with minimal effort. It is used to summarize, sort, organize, group, total, count on average data stored in a database. The table helps in making concise summary out of thousands of rows and columns. Pivot tables can also be referred as a data processing technique that enables a person to arrange and rearrange statistics in order to useful information.

Procedure:

Steps to create pivot table in excel.

- 1) Insert a pivot table
Go to option insert and choose option pivot table & pivot chart we can insert a pivot table in new worksheet / existing worksheet.
- 2) Select the data and table that you want to create as pivot table.

3) choose pivot field.

Select the type of pivot field that you want to insert to your pivot table. The field that you select will be displayed in checkboxes.

4) change position of different fields.

You can change position of different fields in the axis category. Drag desired field to area mentioned below :-

Filters area

column area

row area

value area

After this step, the pivot table will be displayed as a drop down list on the spread sheet.

Conclusion: Here, we have learnt to implement pivot table in excel.

*Prav
18/11/25*

MACRS

Experiment No. 01

(C)

Date: 12/10/25

Rollno: 8019

Working with Macros

Aim: To implement macros in Microsoft word and Microsoft excel

Theory: A macro is a series of commands/instruction grouped together as a single command to accomplish a task automatically. we can use macros in word, and excel. There are different way to create macros, the existing way is to use macros, the existing way is to use macros order.

Procedures:

create a macros in MS word by recording actions.

Steps:

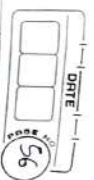
1. In MS excel, click the view tab.
2. click on view below macros → select record macros
3. In record macro window, type a name of new macro in the macro name text field.
4. click: Keyboard icon & assign Key combination in execute menu.
5. In customize keyboard window, click in the press new shortcut field and press combination of keys to use for executing macro.
6. Recording the macro will begin immediately.

Conclusion: Herby, we have learnt, how to work with

macros in MS word & excel.

EXTENSICONS
IN BROWSE?

Experiment No. 08



Date 25/10/25

Roll No : B019

Aim: Installation and use of extensions in browser
various utilities of google like google drive,
google forms, google maps, etc.

Theory: web browsers processes information. Uploads
in the form computer downloads like web, cookie,
all this happens in your browser even visiting a
simple web page requires an exchange of information.
Browser extensions modify the basic flow of
information in some way. An extension is a small
software module you can choose to install to
customize your browser appearance as function.
A few of these software packages come from the
makers of browsers but more often they come
from third party developers. There are thousands
of browser extensions available often / though not
always / through approved sources like chrome /
web store.

Extensions can do almost everything. They might
enable email encryption, ad-blocking, one click
password storage, spell-checking, calendar, add-
ons and on / on. Extensions are specialized
agents working with flow of information through
your browser. They might organize your notes,
protect you from hackers or just transform
how that information appears in browser window.

Examples: utilities of google

google translate: It is a multilingual neural machine translation developed by google to translate documents and websites from one language to another. It offers a website interface, a mobile app for android, IOS and API that helps developers build browser extensions and software applications. How to use google translate extensions.

If you install google translate extensions, you can translate words without having to open a new tab.

1. Go to chrome web store: visit chrome web store with your PC.
2. Search for google translate.
3. Download the extensions. Find google translate on results and click on "Add to chrome".
4. Done. Now you can see extensions icon on browser.

* various utilities of google

a. google drive: It is a free cloud based storage service that enables users to store and access files online. The service syncs stored documents, photos and more of user's devices, tablets and pc's. google drive is used to back up your info and data. It helps you to make your mobile phones space-free. And by logging in your

email-ID by which you have backed up your data in another mobile you can get all the data items in the next device.

b. google maps: google maps is a web device that provides detailed information about geographic regions and sites worldwide. In addition to conversation of road maps, google maps offer serial and satellite view at many locations. In some cities, google maps offers street views comprising photographs taken from vehicles.

c. google forms: google forms is a form creation and data gathering tool developed by google. It's free to use and can be on the web or as an app so can collect responses on any devices. Forms help you build a quest lap pair and collect responses. One of the most common purposes for using google forms is for survey.

Conclusion: learnt the installation and use of extensions in browsers and various utilities of google

Pras
11/11/25

INTERNET EXPLORER

Experiment No. 09

Date: 1/11/25

Rollno: B019

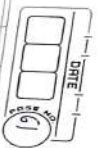
Aim: To use Internet explorer, advance search engines techniques to retrieve information.

Theory:

Search engine : Searching information from electronic digital media is different from print media, when information is stored in electronic digital form, user interface is provided to find relevant information from system. This user interface is a software, which has provisions to accept keywords/terms representing required information to conduct search in format defined in software which has provisional means for searching information from information system and is known as search engine.

Search process: The search process is a set of functions performed for searching relevant information effectively. The process follows some basic steps to conduct, search and get desired result. These steps are as follows.

- i) Recognized and get state the needs.
- ii) Development of search strategy.
- iii) Execution of search strategy.
- iv) Review search results.
- v) Edit search results.
- vi) Evaluation and feedback.



i) Recognize and state the need:
To understand need and purpose of a search

information of a topic may be needed for search knowledge, research and development or any other purpose.

ii) Development of search strategy:
The search string/query is a combination of

information, which represent information. There are aspects namely: syntactic value, semantic value and boolean operators.

a. Syntactic value:

The syntactic value of search string deals with two kinds of formula or connecting symbols through which the keywords or terms are connected to represent the concept searched by the search engines.

b. Semantic value:

The semantic value of search string deals with meaning of string in context of required information in search.

c. Boolean operators:

Boolean operators are simple words (AND, OR, NOT) used as conjunction to combine/exclude keywords in a search.



iii. Execution of search strategy

The searchers should have knowledge of data structure adopted by information system that stores data before executing a search.

a. Field based search :

The search conducted on a particular field of data base to get required information is termed as field based search.

b. Full text search :

It is a searching mechanism which conducts search on each and every field of database and extracts all those records with which match the query.

c. Transaction search :

In a search technique where search is conducted for different forms of word having some common mot.

d. Simple search :

It is a search technique where searcher puts keywords in simple form understanding behaviour of search engine or architecture of database or impact of operators and connect.

* steps to use search engines.

starting your web search

1. go to search engine : A search engine is a

1. Introduction - The purpose of this study is to investigate the relationship between the use of social media and the mental health of young adults. The study aims to explore the impact of social media on self-esteem, anxiety, and depression.

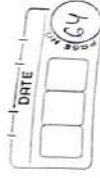
4. Explain the word "mystery".

because you were for better reason to you. I had hope in you. I hope you will be.

224

2. Enter your results by date: The majority of
newer papers have had work done the top of
several pages, including ability to show results
again. A certain way would make the way time
taken in the work much smaller page

The first thing I noticed when I stepped
 out of the car was the smell of
 fresh air. It was a relief after
 being stuck in traffic for so long.
 The sun was shining brightly, and
 the birds were singing. It felt like
 a new beginning.



search engine.

5) Filter your results by type: In general, a regular search displays a list of web links that match what you have typed. But if you are looking others you can click "images", "videos", "news" and other options too.

Conclusion: learnt and implemented how to use internet explorer and advanced search engines

~~11/11/21~~

MOBILE UTILITY

Experiment NO. 10

Date: 8/11/25

Roll no: B019

Aim: To install mobile utility.

Theory: Utility apps are the apps that fulfil functional needs of the user that complete multiple tasks that otherwise would have required physical methods for completion. They can be simple as paying bills or updating work status for employees in field. The functionality of app will vary according to the target audience.

* Mobile app and its features :

CANVA: Canva is an Australian Graphic Design platform that is used to create social media graphics and presentations. The app includes ready made templates for users to use and platform is free and offers paid subscriptions such as Canva Pro and Canva for enterprise for additional functionality. Canva makes creating custom graphics easy. It's easily accessible by phone /tablets and is user friendly.

Not only can you use it to spice up your social media content but also use it to design things like presentation, inputation, posture, brochures, business cards, etc.

You can create and personalize own designs from scratch or choose from thousands of high quality, copy-right free educational templates.

steps to install apps: To download canva app on your laptop first go to canva website. Then click on download button on top right corner of screen. Next select "Download for windows" or "Download for MAC" depending on which type of computer you have. Finally follow instructions to download canva on laptop.

* CANVA features for photos:

Canva for basic cropping and straightening to adding textures and colour correcting, canva's beginner friendly make and colour editing photos simple. You can add textured background to make your photos more dimensional, insert text to create your own meme, create a photogrid, blur your photo and more.

* CANVA offers free video tutorials:

If you are new to graphic design, want to learn about branding or incorporate canva in the virtual classroom, there are free video tutorials on canva with all templates and features available on canva, it feels overwhelming knowing where to begin.

Because mobile devices have become more affordable and portable organizations users have preferred to by them over desktop computers. And with ubiquitous wireless internet access, all varieties of mobile phones are becoming more and more vulnerable to attacks and data breaches. Authentication and authorization across mobile phone offered convenience but increase risk by removing a secured enterprise perimeter's constants. For eg:- a smartphone's capabilities change the way users are rated and authorization is provided locally to devices and applications and services to a network.

* Threats to mobile security

While it is certainly critical to establish and enforce an enterprise wide security policy, a policy alone isn't sufficient to counter the volume and variety of today's mobile threats. In 2019, a study found out that 1 out of 3 of those survey reported a compromise including, mobile device. 47.1% say they suffered down time.

Mobile device security threats may include malicious applications and websites, data leaks, spyware, social engineering attacks and more. They are designed to infiltrate network steal data compromise communications and exploit vulnerabilities found in remote endpoints.

* How to secure mobile devices?

There are many tips on how to keep our devices

- ① Keep your devices locked.
- ② Keep your authentication highest priority.
- ③ update mobile operating system and on-based applications.
- ④ Back up user data on regular basis.
- ⑤ Utilize encryption.
- ⑥ Enable remote data.
- ⑦ Disable bluetooth and wifi scanning when not in use.
- ⑧ Be aware of social engineering techniques.
- ⑨ Be sure not to jail break your devices.
- ⑩ Be sure not to grant unnecessary permissions to applications.
- ⑪ Install mobile security antivirus applications.

conclusion:- In this experiment we performed case study on mobile security and understand its various aspects.

Pratishtha